

Bank Loan Risk Prediction(Project 1)

1.Introduction

This project focuses on analyzing banking customer data in order to identify patterns associated with loan default risk. Financial institutions rely heavily on customer information such as income, age, work experience, home ownership, and marital status to determine whether a customer is likely to repay a loan.

The main objective of the project was to perform Exploratory Data Analysis (EDA), clean and prepare the data for machine learning, and develop a Logistic Regression model capable of predicting risky customers. Through visualization and statistical analysis, important business insights were identified to assist the banking institution in reducing loan default rates and improving decision-making.

The project also aimed to evaluate the performance of the machine learning model using accuracy score, confusion matrix, classification report, and ROC curve analysis.

2.Business Problem

The banking institution has experienced a high rate of loan defaults among personal loan customers. This creates financial losses and affects operational efficiency within the loans department. Traditional manual methods of assessing customer risk may not be sufficient for accurately identifying risky customers.

The business problem addressed in this project was therefore to develop a predictive system capable of identifying customers who are more likely to default on loans. By predicting loan risk early, the bank can improve loan approval decisions, minimize financial losses, and strengthen risk management processes.

3.EDA

Several visualizations were developed during the exploratory data analysis process to better understand the dataset and identify important trends. A count plot was used to analyze the distribution of risky and non-risky customers. The visualization revealed that the majority of customers were classified as non-risky, although a significant number of risky customers were also identified.

Histograms were used to analyze the distributions of income, age, and experience. The analysis showed that income and customer experience varied widely across the dataset. Boxplots and countplots were also used to compare customer characteristics such as marital status, home ownership, and car ownership against loan risk.

A correlation heatmap was developed to identify relationships among numerical variables. The heatmap helped determine which factors had stronger relationships with loan risk and provided useful insights for machine learning preparation.

4.Data cleaning

Data cleaning was performed to prepare the dataset for machine learning analysis. Duplicate records were removed to improve data quality. Missing values were identified and handled using median value imputation for numerical columns.

Categorical variables such as marital status, profession, home ownership, city, and state were converted into numerical values using Label Encoding. This step was necessary because machine learning algorithms cannot process text-based categorical data directly.

The dataset was then divided into independent variables (features) and the dependent variable (Risk_Flag). Finally, the dataset was split into training and testing sets to evaluate the performance of the machine learning model.

5.Machine learning model

A Logistic Regression model was developed to predict customer loan risk. Logistic Regression was selected because it is effective for binary classification problems where the target variable contains two possible outcomes, namely risky and non-risky customers.

The model was trained using the training dataset and tested using unseen test data. Predictions were generated and evaluated using several performance metrics including accuracy score, confusion matrix, classification report, and ROC curve analysis.

6.Findings

The machine learning model achieved an accuracy of 88%

The evaluation results demonstrated that the Logistic Regression model performed effectively in predicting customer loan risk. The confusion matrix showed that the model correctly classified a large number of both risky and non-risky customers.

The classification report further demonstrated strong precision, recall, and F1-scores, indicating reliable prediction capability. The ROC curve also showed that the model had good discriminatory power in distinguishing between risky and non-risky customers.

7.Conclusions

This project successfully demonstrated the use of data analytics and machine learning techniques in solving banking risk management problems. Exploratory data analysis provided useful insights into customer characteristics and financial behavior patterns.

The Logistic Regression model achieved strong predictive performance and can assist the banking institution in improving loan approval decisions and minimizing default rates. The project highlights the importance of data-driven decision-making in modern banking operations.

8. Recommendations

The banking institution should continue investing in data analytics and machine learning technologies to improve risk assessment processes. Additional customer variables such as credit history and transaction behavior may further improve prediction accuracy.

The institution should also regularly update and retrain the machine learning model using new customer data to maintain high predictive performance over time. Furthermore, customers identified as high-risk should undergo additional financial assessment before loan approval.

9.Graphs



